

Business Report

Sales Performance Analysis Report

Junior Data Analyst Portfolio Project

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Executive Summary

Introduction

This project analyses retail sales performance using transactional sales data from the Sample Superstore dataset. The objective was to identify trends in sales, profitability, customer behaviour, product performance, and discount effectiveness to support data-driven business decisions.

Interactive dashboards were developed in both Microsoft Excel and Power BI to transform raw data into meaningful business insights. The analysis focused on identifying the strongest-performing regions, products, customer segments, and categories while investigating how discounts influence profitability.

The findings provide practical recommendations that could help improve profitability and optimise future business strategies.

Business Problem

Retail organisations generate large volumes of sales data every day. However, without effective analysis, it is difficult to understand which products, customers, and regions contribute most to business performance.

This project was designed to answer several key business questions and demonstrate how business intelligence tools can support decision-making through data analysis and visualisation.

Project Objectives

The objectives of this project were to:

- Analyse retail sales performance across different regions.
- Identify the highest-performing products and categories.
- Evaluate customer segment profitability.
- Investigate the relationship between discounts and profit.
- Build interactive dashboards using Microsoft Excel and Power BI.
- Produce clear business recommendations based on analytical findings.

Dataset Overview

The analysis uses the Sample Superstore dataset from Kaggle, which contains approximately 10,000 retail transactions.

The dataset includes information relating to:

- Sales
- Profit
- Quantity Sold
- Discount
- Region
- Customer Segment
- Category
- Sub-Category

Before analysis, the dataset was reviewed to ensure data types were correct and suitable for reporting.

Methodology

The project followed a structured analytical process:

Step 1: Data Preparation

The dataset was imported into Microsoft Excel and Power BI Desktop.

The data was reviewed to confirm:

- Correct data types
- No missing key fields
- Consistent categorical values
- Valid numerical values

Step 2 – KPI Development

Four key performance indicators (KPIs) were calculated.

Table 1: KPI Development

KPI	Value
Total Sales	£2,296,958
Total Profit	£286,324
Profit Margin	12.47%
Quantity Sold	37,871

Step 3 – Dashboard Development

Two dashboards were developed.

Table 2: Excel Dashboard

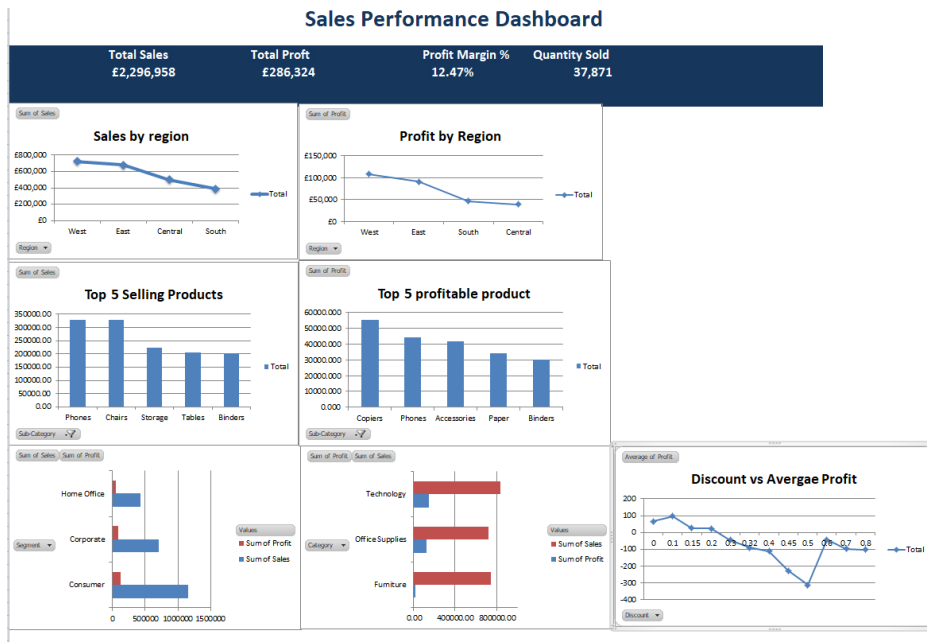
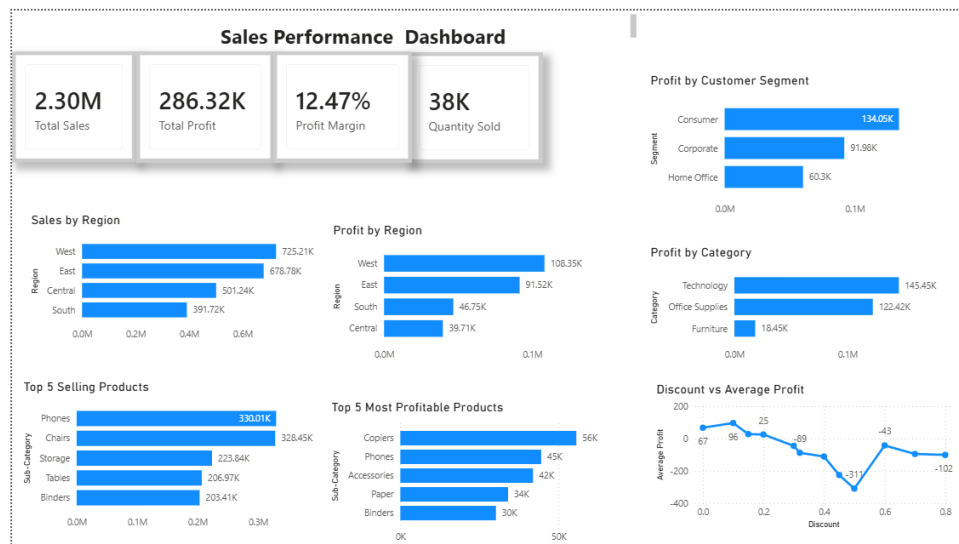


Table 3: Power BI Dashboard



Business Questions

The analysis addressed the following business questions:

1. Which region generated the highest sales?
2. Which region generated the highest profit?
3. Which products generated the highest revenue?
4. Which products generated the highest profit?
5. Which customer segment contributed the most profit?
6. Which product category was the most profitable?
7. Do larger discounts reduce profitability?

Analysis & Findings

Regional Performance

As shown in Table 4 the West region generated the highest overall sales and profit. The East region also performed strongly, while the South generated the lowest overall sales.

These findings suggest that the West region is the strongest-performing market within the business.

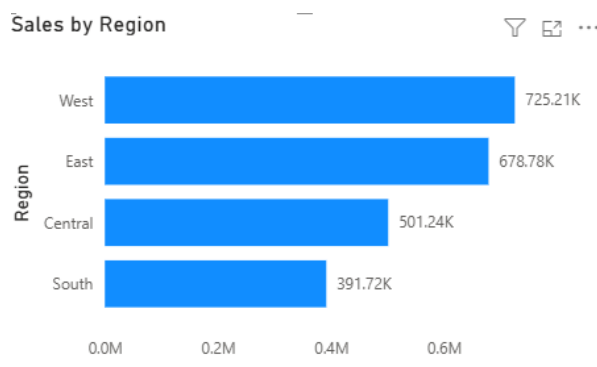


Table 4: Sales by Region

Product Performance

Table 5 shows phones generated the highest sales revenue, making them the strongest-selling product. However, Copiers generated the highest overall profit, indicating a stronger profit margin despite lower sales volume.

This demonstrates that the highest-selling products are not always the most profitable.

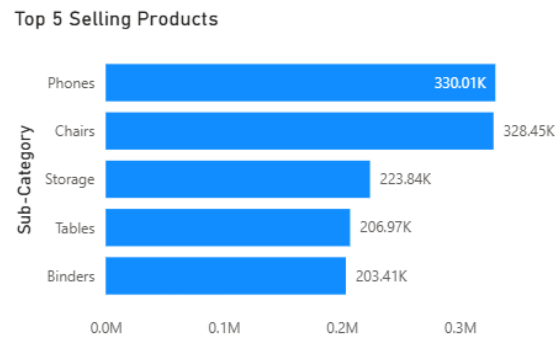


Table 5: Top 5 Selling Products

Customer Segment Analysis

The Consumer segment generated the highest total profit which can be seen through table 6.

Although Home Office customers generated lower overall profit, they achieved a comparatively stronger profit margin, suggesting potential opportunities for future business growth.

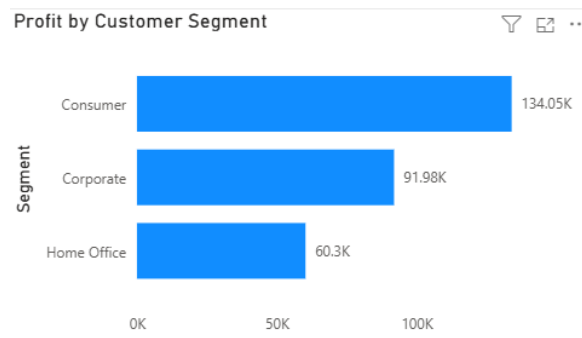


Table 6: Profit by Customer Segment

Category Performance

According to table 7 technologies produced the highest overall profit. Office Supplies also performed well. Furniture generated substantial sales but comparatively low profit.

Earlier analysis identified Tables and Bookcases as loss-making products, explaining the weaker performance of the Furniture category.

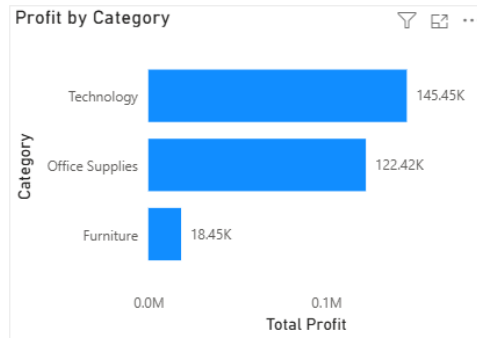


Table 7: Profit by Category

Discount Analysis

The analysis explored how increasing discount levels affected profitability.

Table 8 demonstrates that the average profit remained positive at lower discount levels. However, discounts above approximately 30% were associated with negative average profit.

Although this relationship does not necessarily imply causation, it indicates that aggressive discounting may reduce profitability and should be monitored carefully.

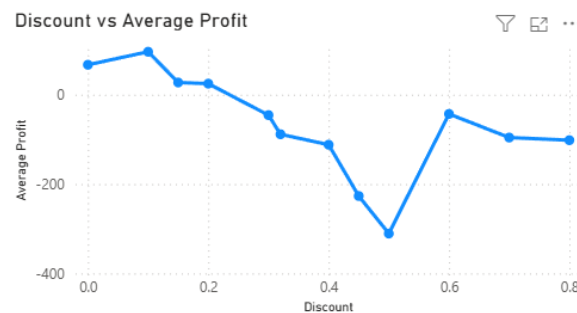


Table 8: Discount Vs Average Profit

Key Insights

The analysis produced several important business insights:

- The West region consistently generated the strongest business performance.
- Technology products delivered the highest profitability.
- Phones generated the highest sales revenue.
- Copiers generated the highest profit.
- Furniture underperformed despite high sales.
- Tables and Bookcases reduced Furniture profitability.
- Discounts above approximately 30% were associated with negative average profit.

Business Recommendations

Based on the findings, the following recommendations are proposed:

Recommendation 1

Continue investing in Technology products due to their consistently high profitability.

Recommendation 2

Review pricing strategies for Tables and Bookcases to improve Furniture profitability.

Recommendation 3

Monitor discount strategies carefully and avoid excessive discounting where possible unless supported by a clear commercial objective.

Recommendation 4

Continue strengthening the Consumer segment while exploring opportunities to increase engagement with the profitable Home Office segment.

Recommendation 5

Maintain investment in high-performing regions, particularly the West, while investigating opportunities to improve sales and profitability in the South region.

Conclusion

This project demonstrates how Microsoft Excel and Power BI can be used together to transform transactional retail data into meaningful business intelligence.

Through dashboard development, KPI reporting, and business analysis, the project identified key trends in regional performance, product profitability, customer behaviour, and discount effectiveness.

The recommendations presented in this report provide practical actions that could help improve profitability and support future business decision-making.

References

- Microsoft. Power BI Documentation.
- Microsoft. Excel Documentation.
- Sample Superstore Dataset (Kaggle).